hydrogen has not picked up—despite the push to build a supply ecosystem—is cost, which is way more than the price of conventional grey hydrogen. Policy support needed to create assured demand is inadequate as well. The result: ₹17,490 crore worth of incentives under SIGHT, out of the mission's outlay of ₹19,744-crore, still unclaimed, and an industry waiting for a clearer market signal.

AMBITION V/S REALITY

Notwithstanding India's ambition to build a green hydrogen economy, India Inc is not fully on board. Larsen and Toubro (L&T) has put on hold plans to expand the electrolyser manufacturing capacity at its Hazira plant in Gujarat from 400 MW to 1 GW.

Industry sources told *Business Today*, on condition of anonymity, that low demand has forced big players such as Reliance Industries and Adani Group, which had announced 3-GW and 5-GW electrolyser plants, respectively, to slow expansion plans. According to Nishaanth Balashanmugam, CEO and Director of the industry body, Green Hydrogen India (GH2 India), this is a "sector-wide issue".

Reliance had announced its electrolyser plant in August 2024. It was expected to be operational by the end of 2026. A company spokesperson says on condition of anonymity the plant is "under construction" with no clarity on the production schedule. It had also announced plans to produce 3 million tonnes (MT) of green hydrogen and derivatives (green ammonia) by 2032, work on which will commence in 2028 or 2029, adds the spokesperson. The group's electrolysers are meant for internal and external use, while green hydrogen is for captive use. "As our Chairman, Mukesh Ambani, has said, green hydrogen will become viable only when its price falls below $1 per kg," says the spokesperson. At present, it costs $3.5-5 per kg without the SIGHT subsidy and $3-3.75 with it.

The Adani Group announced plans to build the electrolyser plant in 2022 and set up a pilot project in June 2025. However, the company did not clarify its production schedule. The spokesperson for the group has not responded to *Business Today*'s queries.

The demand constraint has been flagged for quite a while. In November 2025, the global think tank, In-

BIG PLANS, SMALL SUCCESS

1. A TARGET OF FIVE MILLION TONNES OF GREEN HYDROGEN BY 2030
2. EXPERTS SAY UNCLAIMED INCENTIVES SINCE 2023 AT ₹17,490 CRORE
3. ON SEPTEMBER 7, L&T PUT ON HOLD ITS PLAN TO EXPAND ELECTROLYSER MANUFACTURING CAPACITY AT ITS HAZIRA PLANT IN GUJARAT FROM 400 MW TO 1 GW
4. ADANI GROUP HAD ANNOUNCED A FIVE GW ELECTROLYSER PLANT IN 2022. IT SET UP A PILOT PROJECT IN JUNE 2025 WITH NO CLARITY ON ITS PRODUCTION SCHEDULE
5. IEEFA, IN A REPORT, FLAGGED THAT 94% OF THE PLANNED CAPACITY IS YET TO MOVE BEYOND THE ANNOUNCEMENT STAGE, AND ONLY 2.8% IS OPERATIONAL

COMPARATIVE COST OF GREY AND GREEN HYDROGEN

TYPE	COST PER KG
▸ Grey hydrogen	$1.9-2.5 (₹160)
▸ Green hydrogen	$3.5-5 (₹295-420)
▸ Green hydrogen with SIGHT incentives	$3-3.75 (₹260-310)
▸ Industry breakeven for uptake	Less than $2 (₹170)

NOTE SIGHT IS STRATEGIC INTERVENTIONS FOR GREEN HYDROGEN TRANSITION **SOURCE** CRF